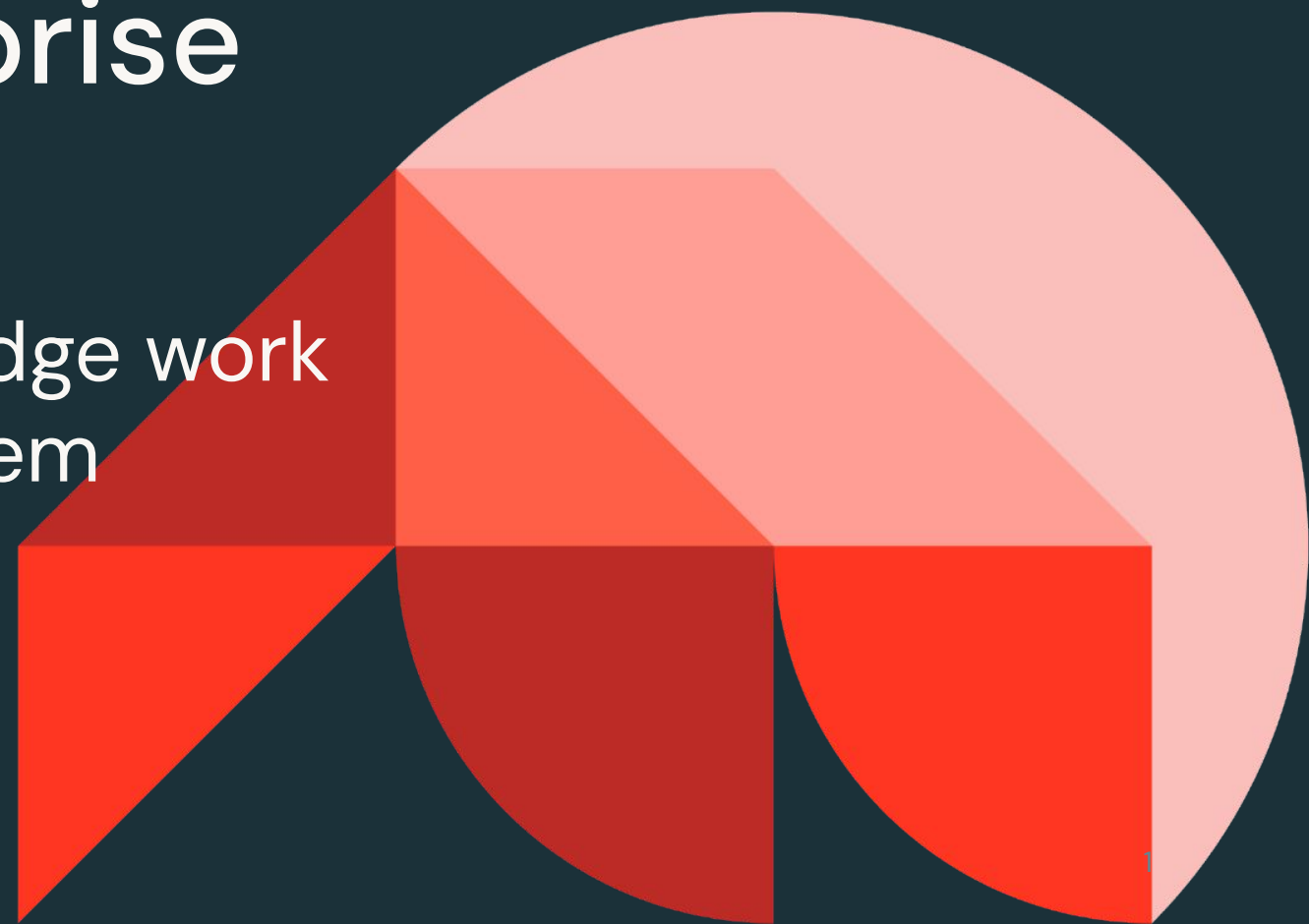




Building an enterprise “truth registry”

The foundation for knowledge work
on an open operating system



Agenda

1 AI

2 AI-ready context across domains

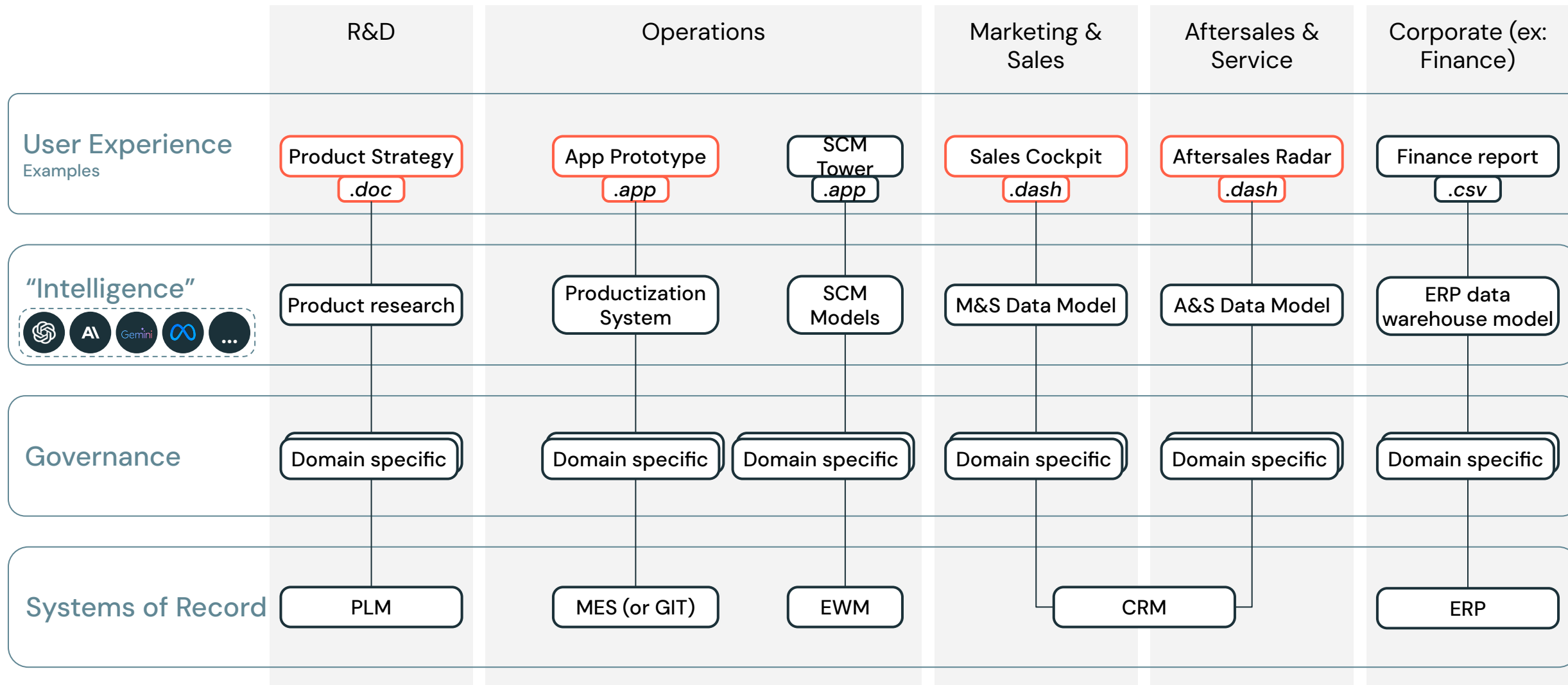
3 Open and extensible governance

4 Next Steps

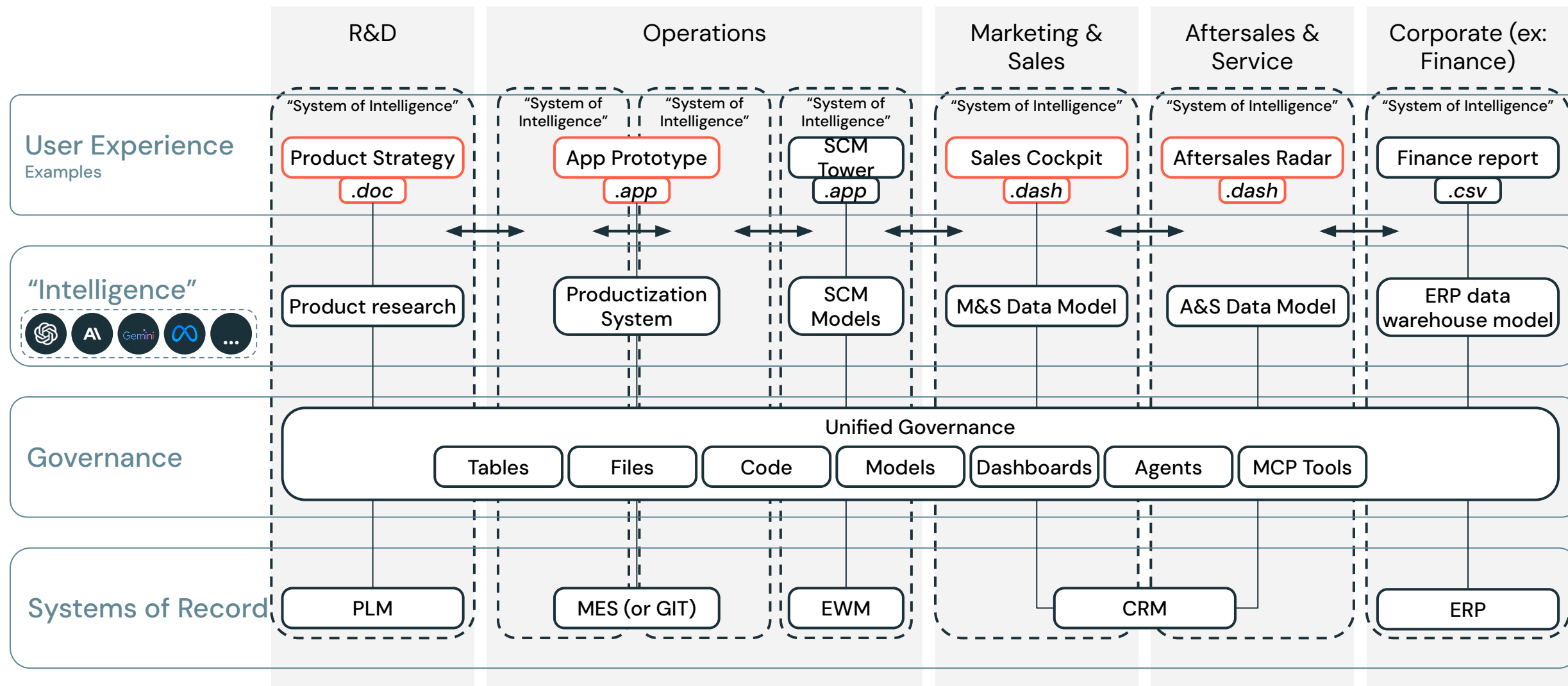


**Knowledge work needs information
(data + context)
All rooted on trust**

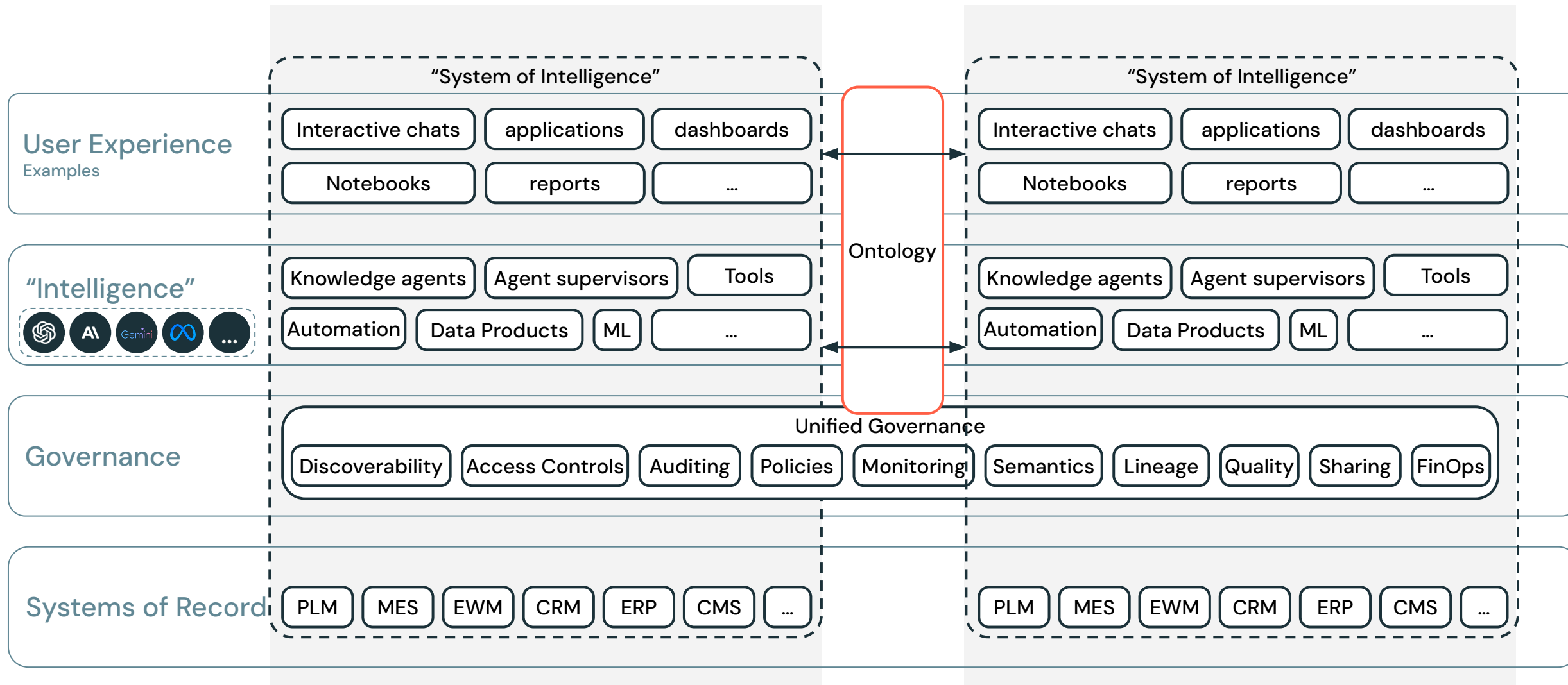
Data & AI: Systems of Intelligence



Data & AI: Systems of Intelligence



Data & AI: Systems of Intelligence



Agenda

1

Data & AI: Systems of Intelligence

2

AI-ready context across domains

3

Open and extensible governance

4

Next Steps

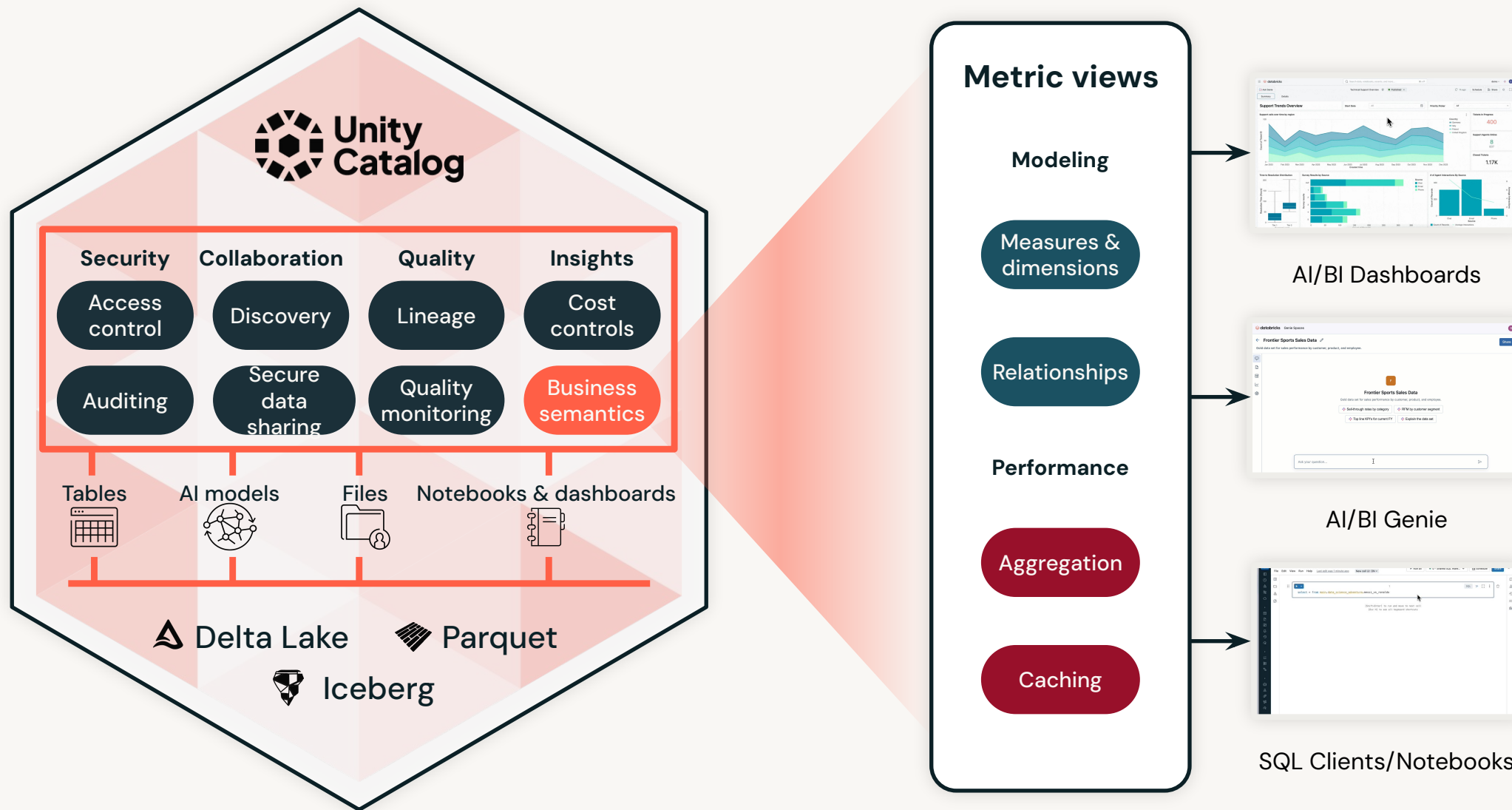


Semantic maturity

Moving from esoteric comments to true semantic consistency

Unified, business facing metrics	• Single source of truth for metrics and their lifecycle, accessible from all BI tools • Terminology that is understandable by decision makers • Semantic layer can optimise performance of queries based on usage • Governance on <i>business logic</i>, not just data
Basic Documentation	• Tables and column comments • Separate business glossary • Specialist help needed to find and access data • Low discoverability and reuse

Unity Catalog as your semantic layer



Semantic maturity

Moving from esoteric comments to true semantic consistency

Domain-specific ontology	• Knowledge graph captures semantic meaning of data & relationships • GenAI can use ontology to aid discoverability and automate insights • Organisation-wide view of concepts enhances alignment on terminology • Interoperability between teams and with industry standard definitions of concepts and processes
Unified, business facing metrics	• Single source of truth for metrics and their lifecycle, accessible from all BI tools • Terminology that is understandable by decision makers • Semantic layer can optimise performance of queries based on usage • Governance on <i>business logic</i>, not just data
Basic Documentation	• Tables and column comments • Separate business glossary • Specialist help needed to find and access data • Low discoverability and reuse

databricks | What is Project Ontos? (git)

Apply the RDF-defined ontology to Unity Catalog objects to **enhance semantic enrichment**.

AI-driven ontology application automatically enhances Unity Catalog semantic integrity.

Data Discovery via Business Concept Alignment

ABAC enforces security protocols based on semantic analysis.

It clarifies whether two datasets represent the same concept, detecting semantic inconsistencies.



Ontos (Custom App)

Machine Readable Ontology & Semantic

Connect data to the concepts it represents



AI/BI

Agentic business intelligence

Data Products

Business Glossary

Ontology

Data Contracts

Catalog

Build on Open Standards

RDF

ODCS

ODPS

Unity Catalog



Agent Bricks

Agentic AI & Machine Learning



Unity Catalog



DELTA LAKE

ICEBERG



Postgres



Agenda

1

Data & AI: Systems of Intelligence

2

AI-ready context across domains

3

Open and extensible governance

4

Next Steps



databricks



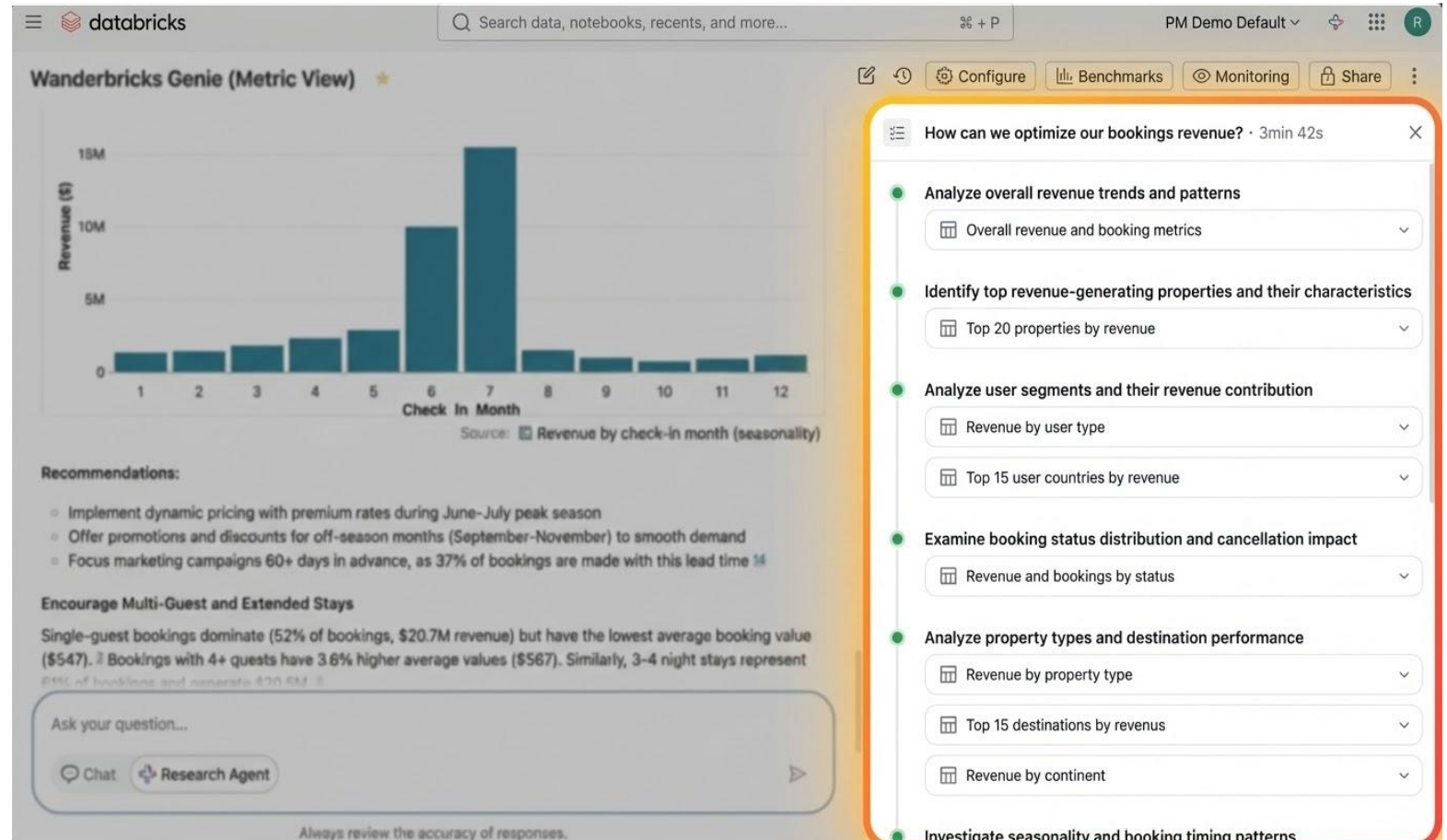
Move From Plausible to Provable

The leap from probabilistic pattern matching to deterministic reasoning required for enterprise trust

Semantic(Ontology) governed reasoning: Agents response are governed through semantic enriched from ontology.

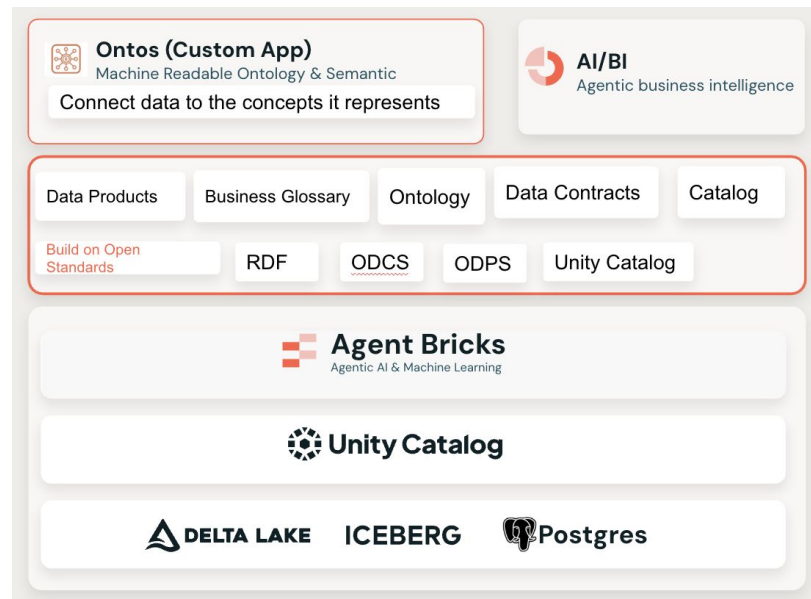
Agents checks semantic(Ontology) before access.

User Trust the GenAI response.

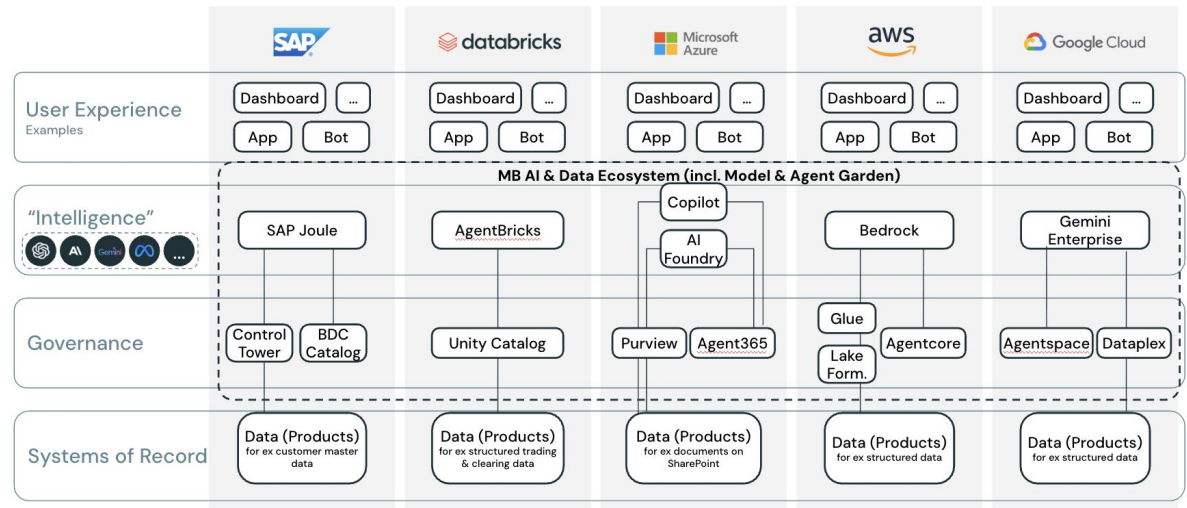


Open and Extensible Governance

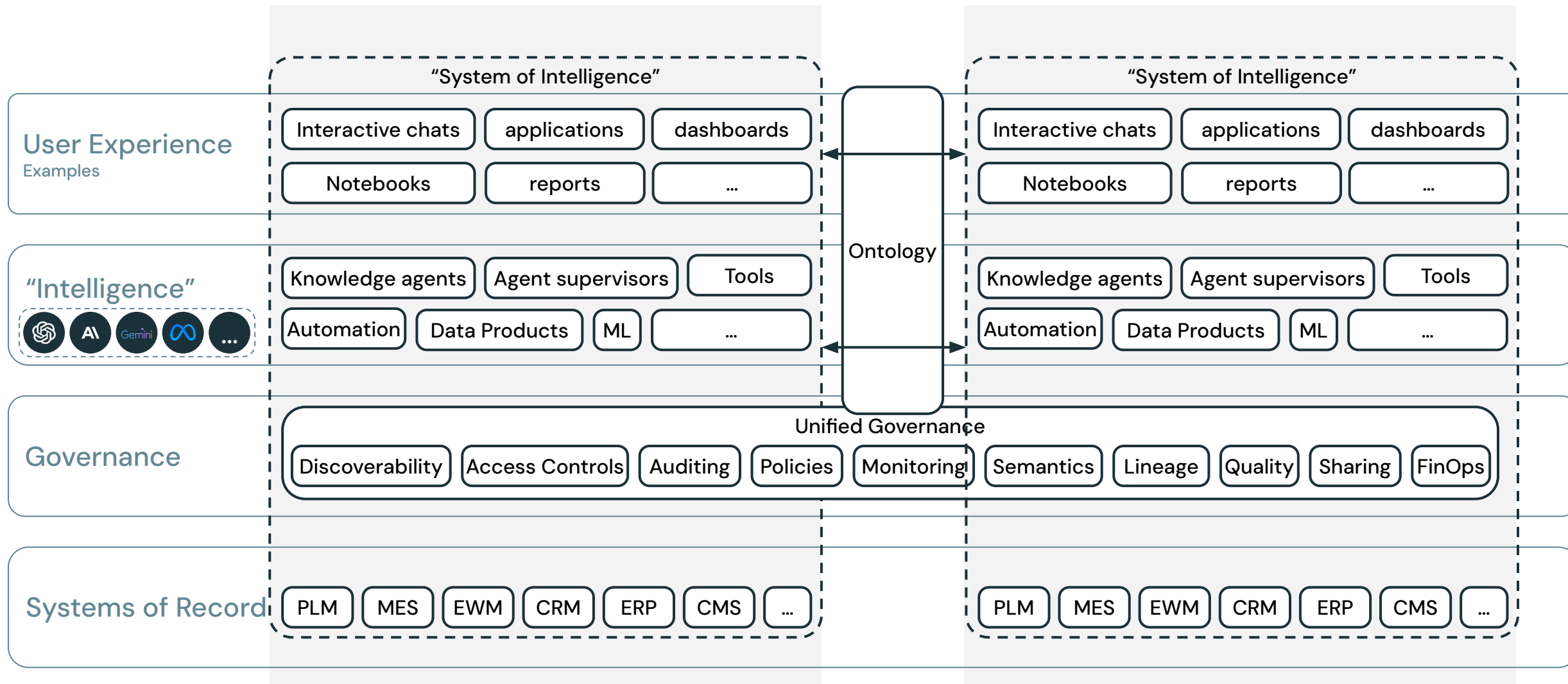
1. **Data products** are the foundation for **competitive advantage**
2. **Enable stakeholders** to follow **data gravity** within tech stack guardrails
3. Ensure **interoperability, traceability** and **governance** across the group



Open and Extensible Governance

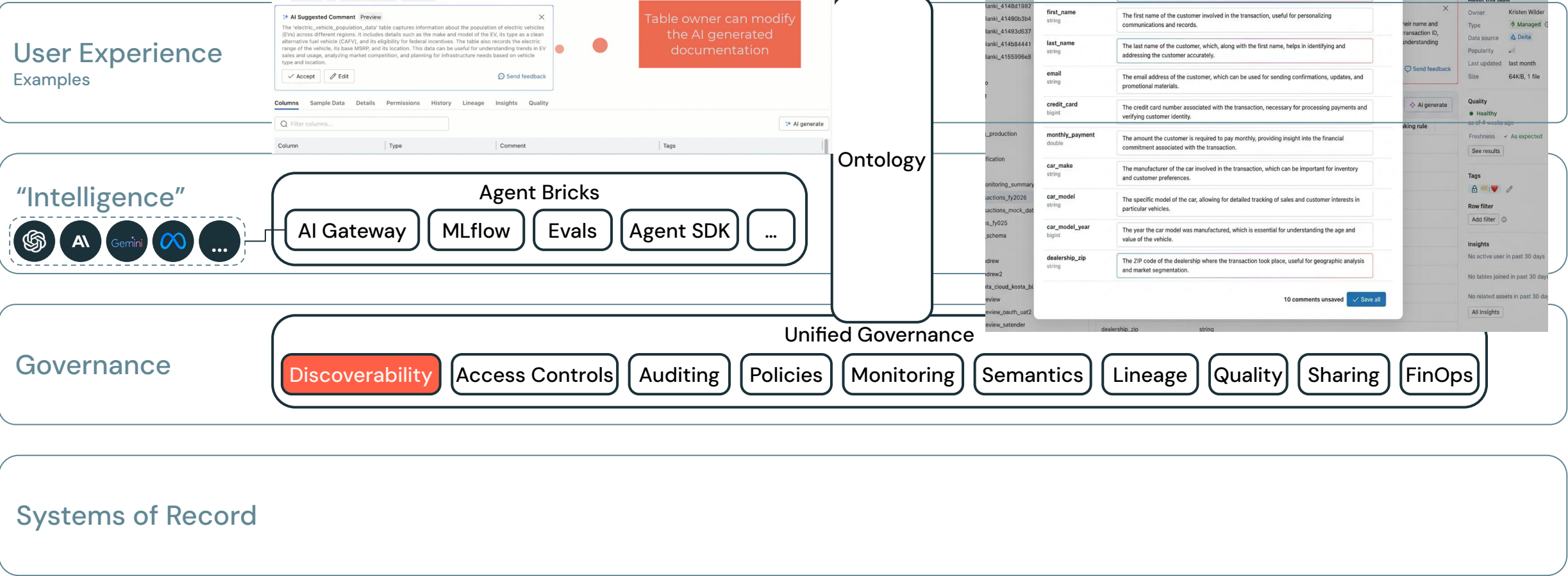


Open and Extensible Governance



Open and Extensible Governance

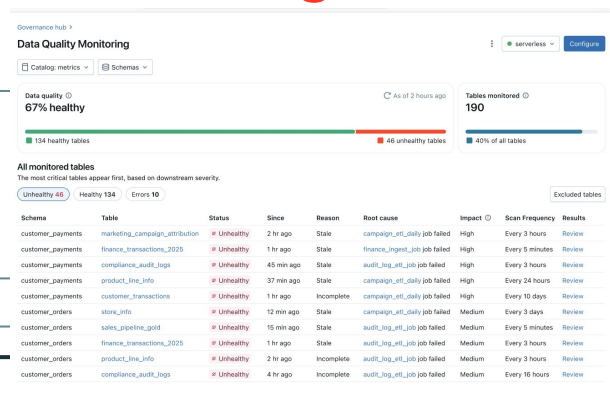
Data Quality with AI Blog



Open and Extensible Governance

Data Quality with AI Blog

User Experience Examples



"Intelligence"



AI Gateway

MLflow

Evals

Agent SDK

...

Ontology

Governance

Unified Governance

Discoverability

Access Controls

Auditing

Policies

Monitoring

Semantics

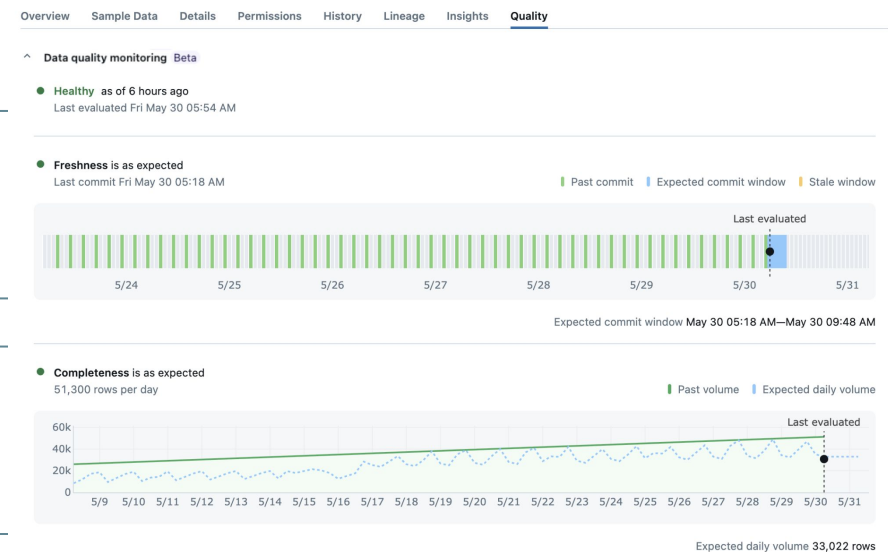
Lineage

Quality

Sharing

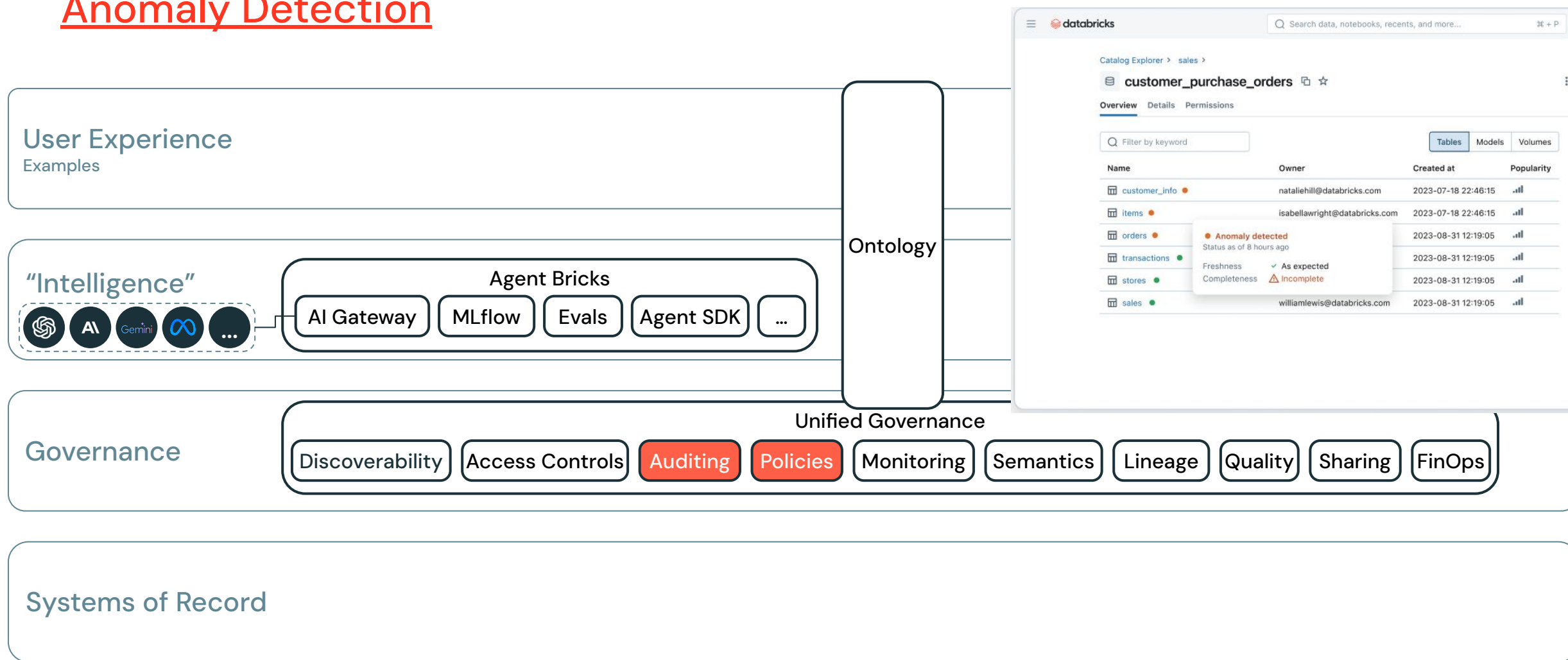
FinOps

Systems of Record



Open and Extensible Governance

Anomaly Detection



Open and Extensible Governance

Ontos on Marketplace

User Experience Examples

"Intelligence"



Agent Bricks

AI Gateway

MLflow

Evals

Agent SDK

...

Ontology

Governance

Unified Governance

Discoverability

Access Controls

Auditing

Policies

Monitoring

Semantics

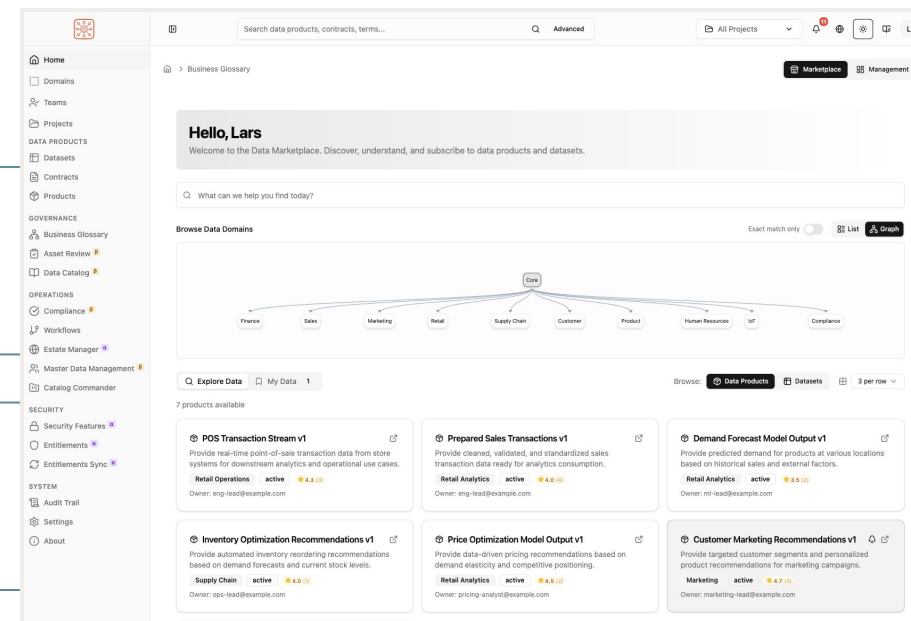
Lineage

Quality

Sharing

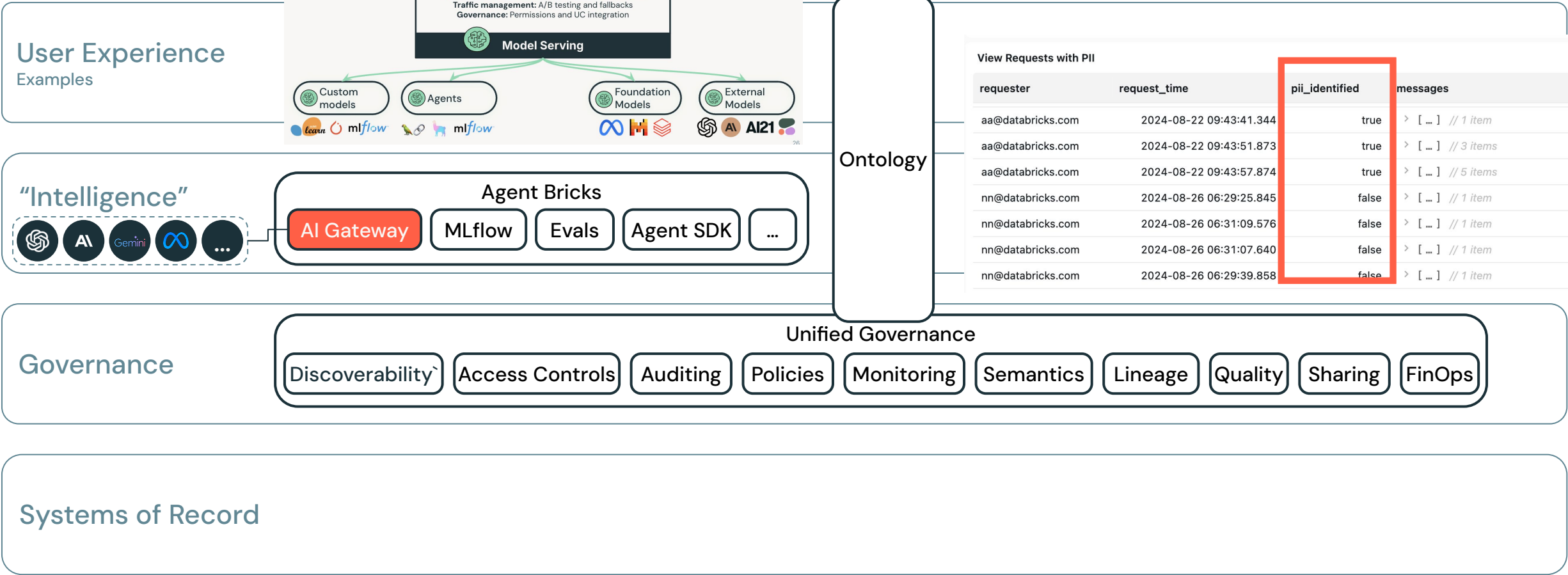
FinOps

Systems of Record



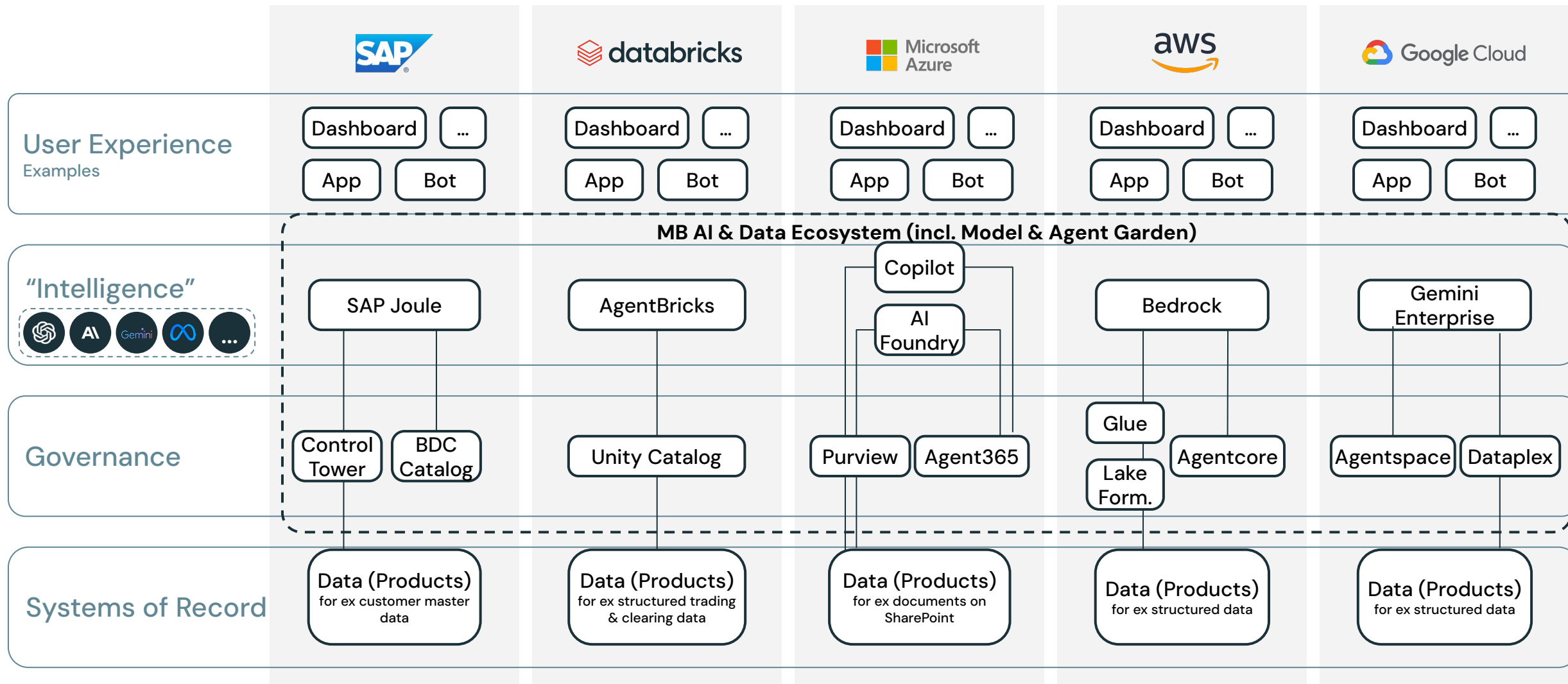
Open and Extensible Governance

AI Gateway



Open and Extensible Governance

Motzvorlage



Agenda

1

Data & AI: Systems of Intelligence

2

AI-ready context across domains

3

Open and extensible governance

4

Next Steps



databricks





Building Talk-to-Data Semantic Layer at Scale

Mercedes-Benz Korea Success Story

Challenge

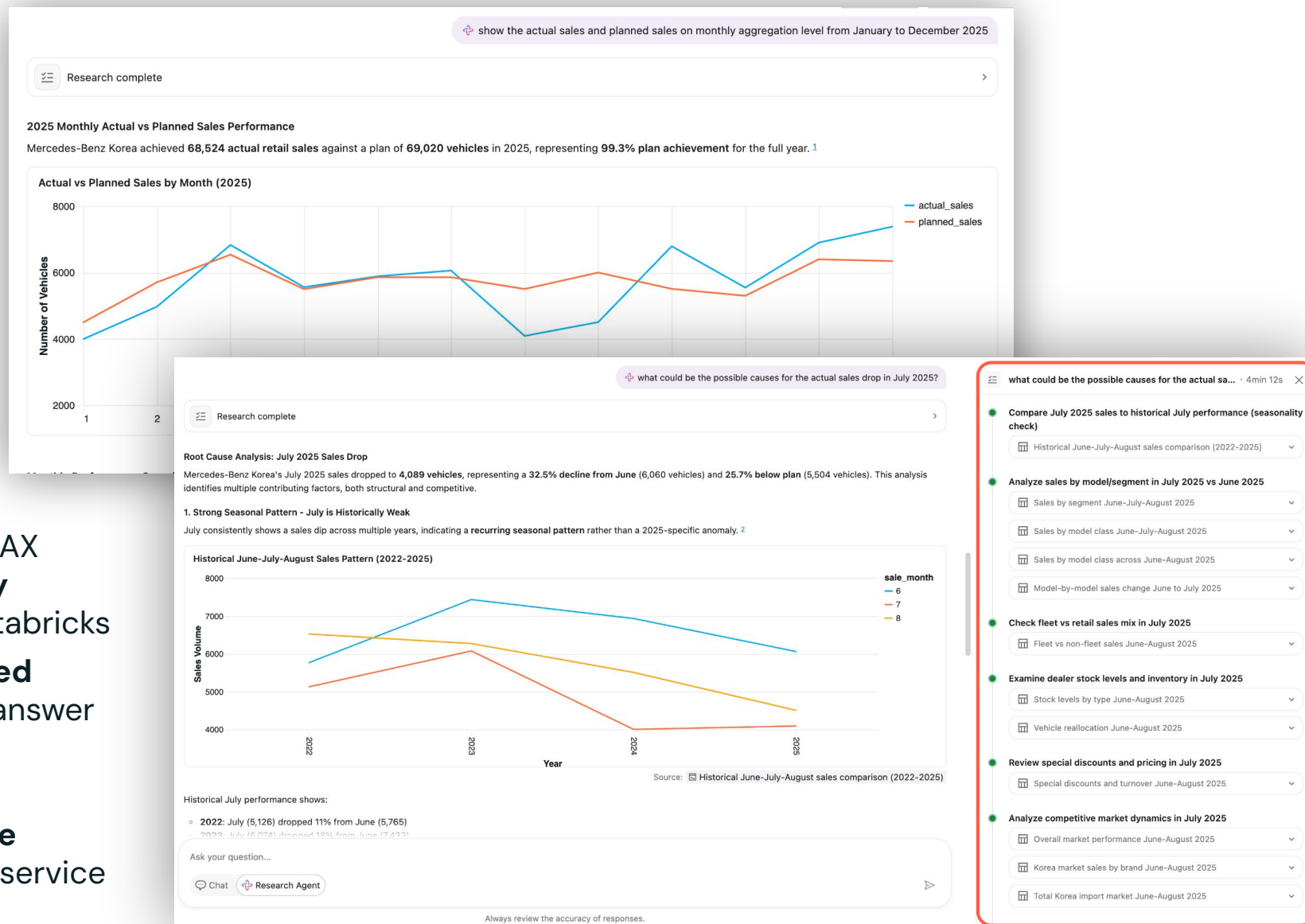
- Sales KPIs maintained in complex Power BI semantic models and a legacy Master KPI catalog
- Difficult to reuse KPI logic and **scale with consistency** for AI agents and across platforms

Solution

- Automated conversion of Power BI DAX measures to **AI-ready and centrally governed** Business Semantics in Databricks
- Genie understands and uses **enriched semantic metadata** to reason and answer

Impact

- **Accurate, consistent, and scalable** talk-to-data experience for AI self-service across global sales markets



Knowledge work needs “truth”



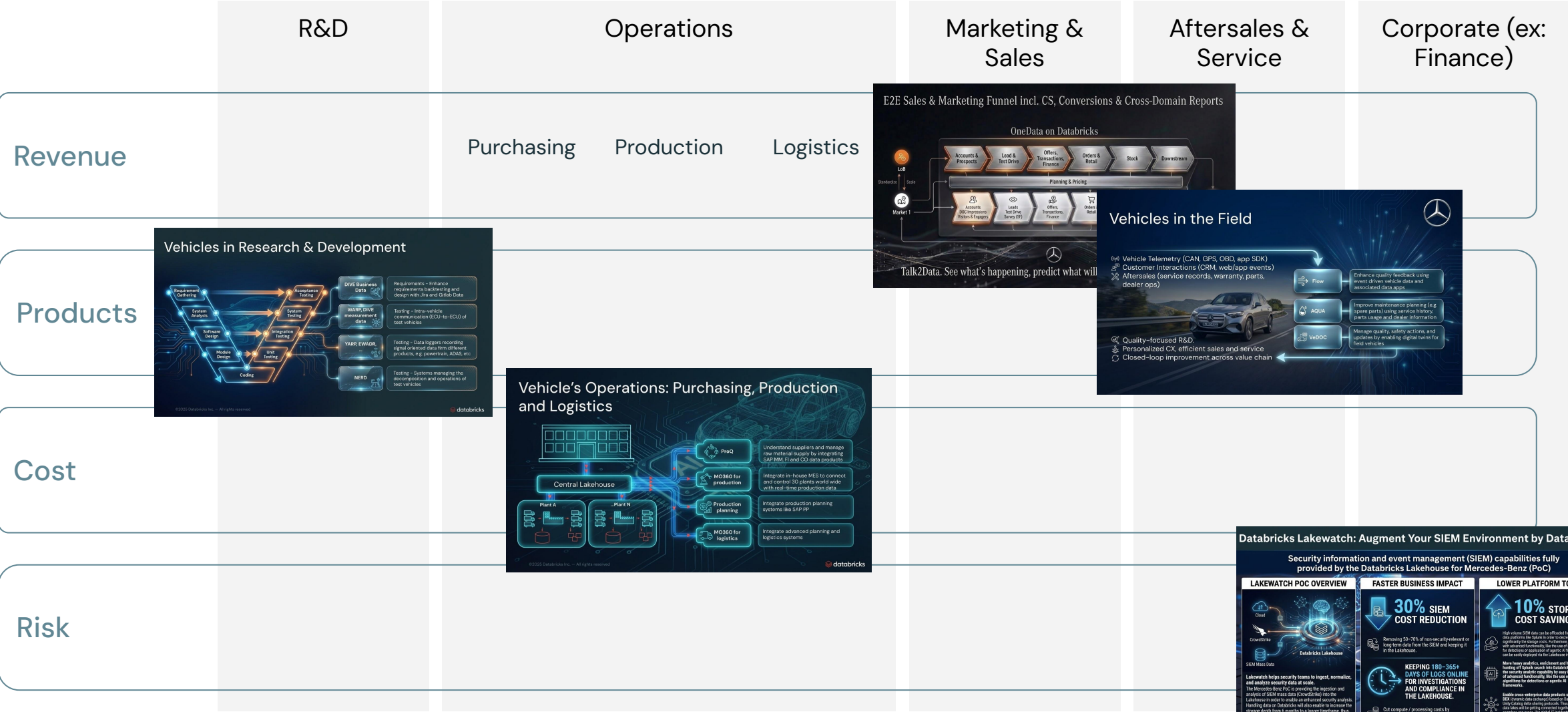
“We are the biggest small company in the world” and the “gold mine” analogy

– Jensen Huang, Founder and CEO of NVIDIA
with Ali Ghodsi, Co-founder and CEO of Databricks (June 2024)

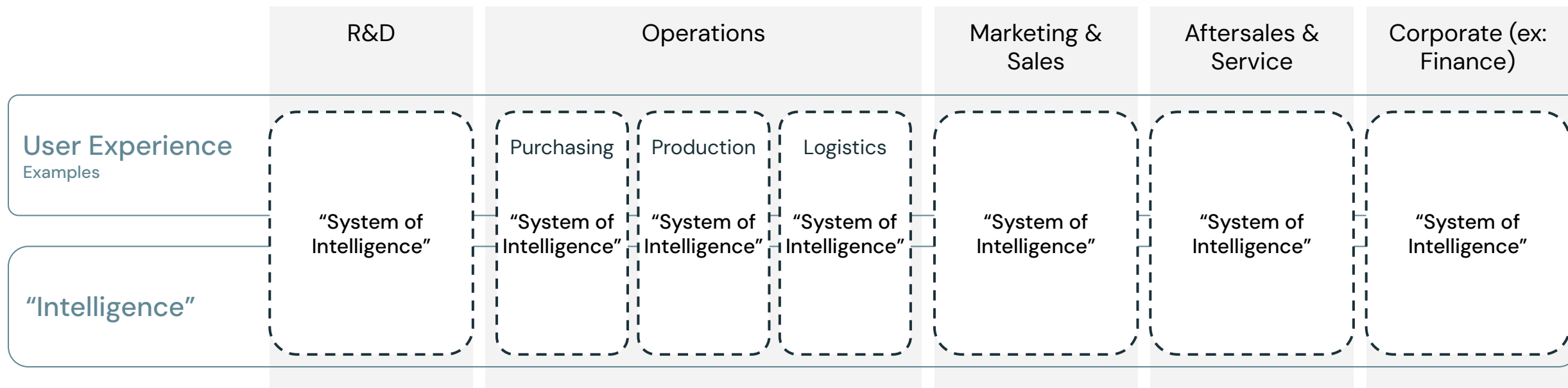
*“Only **12% of CIOs** say their organizations are highly successful in delivering on their data strategy and just **2% of executives** say their organizations are achieving strong business results from AI”*

MIT—Building a high-performance data and AI organization—July 2025

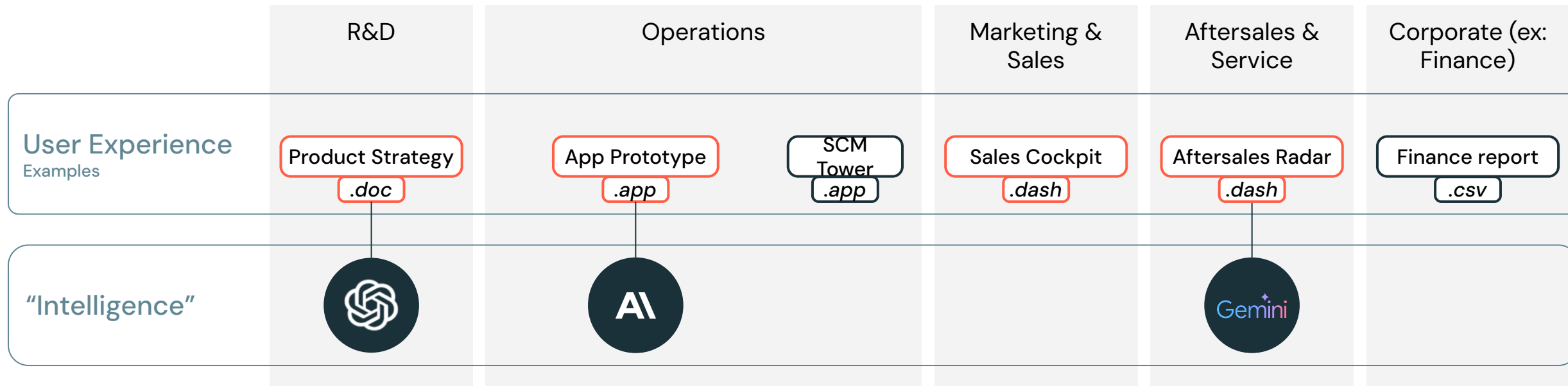
Data & AI: Change along the Value Chain



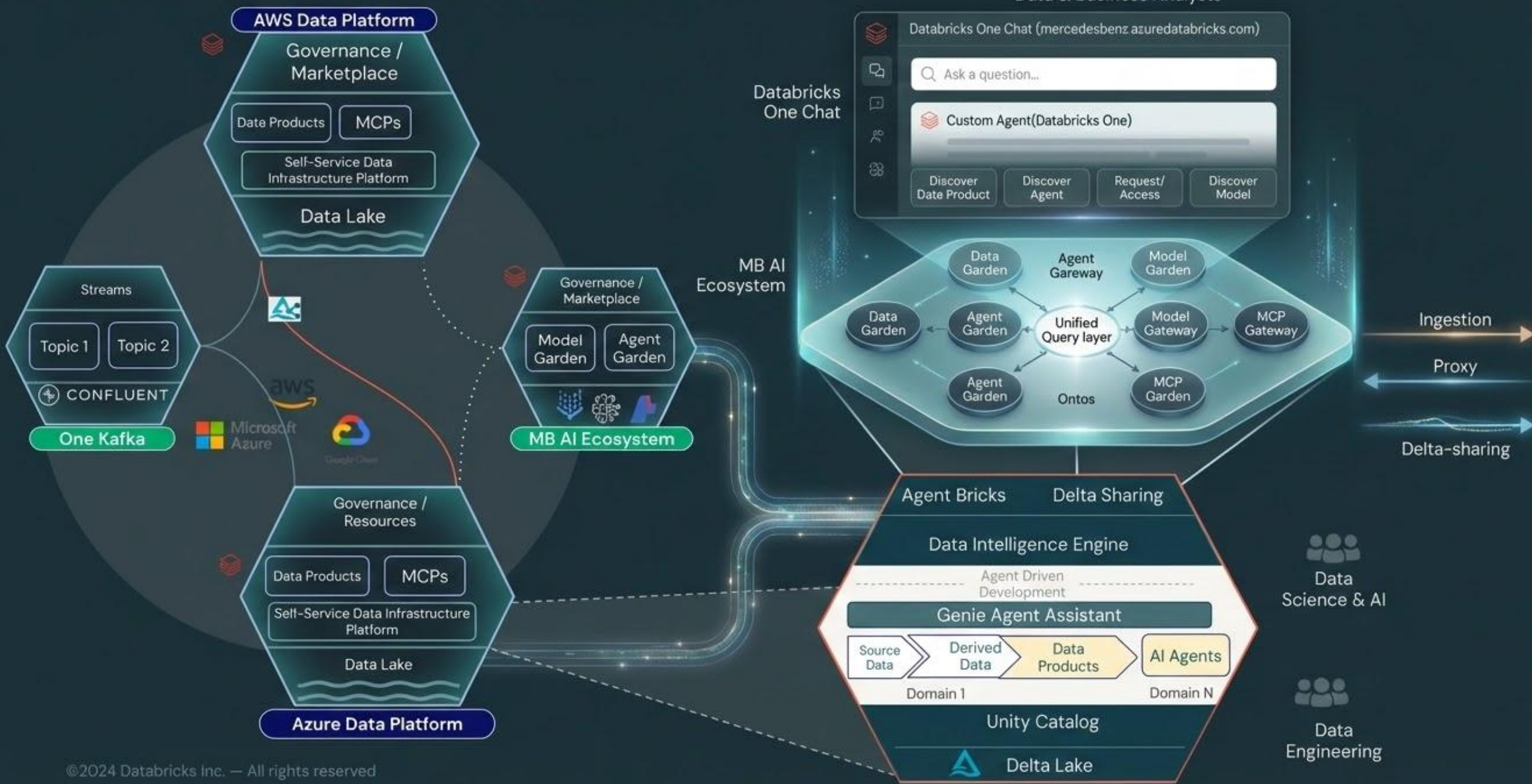
Data & AI: Systems of Intelligence



Data & AI: Systems of Intelligence



Data & AI Self-Service



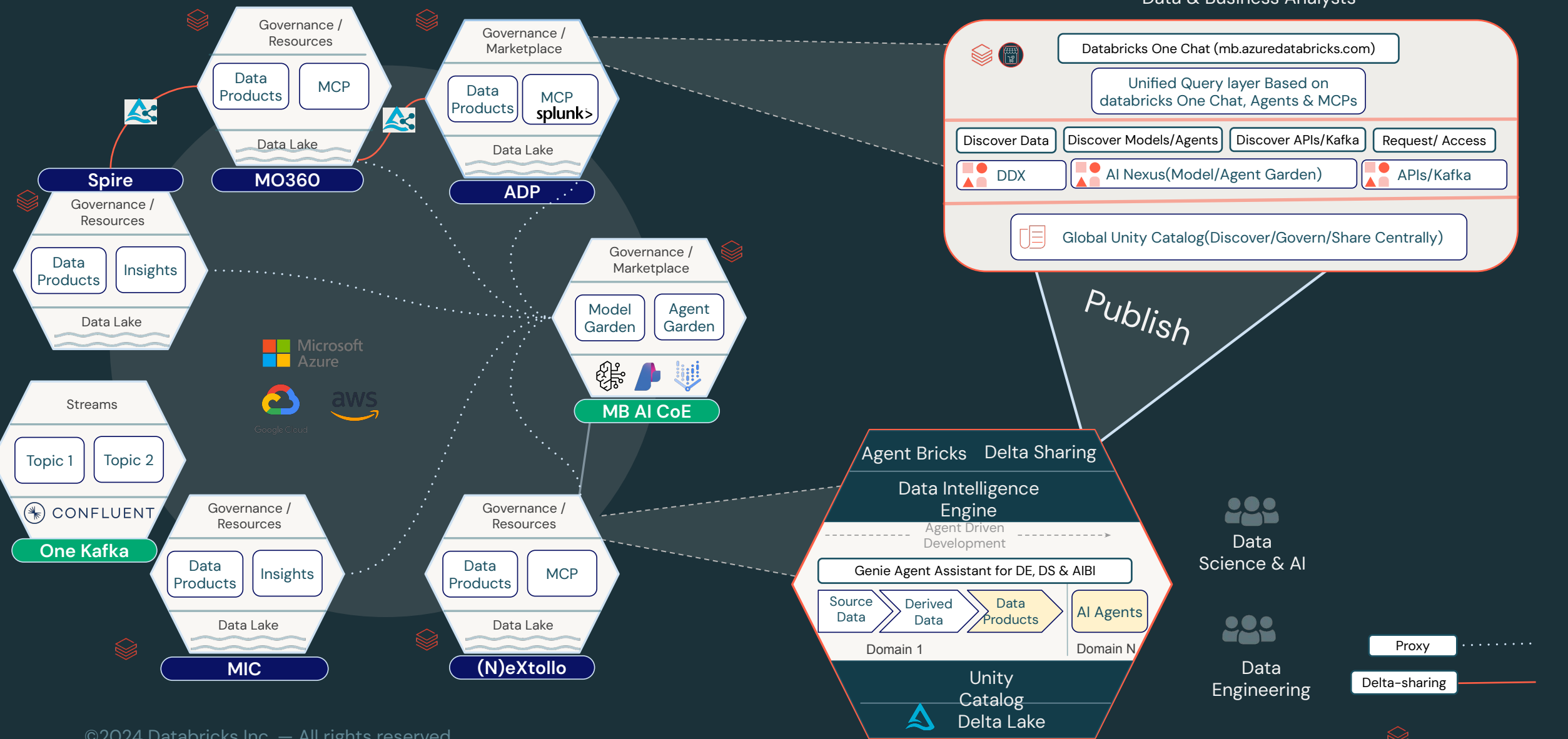
Accelerate your data strategy: Build the Lakehouse along the value chain of products



Data & AI Self-Service(Marketplace)



Data & Business Analysts



Every industry will benefit from AI



Financial Services

Market Intelligence for Alpha Generation

Personalized Offers for Cross-sell / Up-sell

Financial Crime Detection & Remediation

Broker Decision Support to Optimize Premiums

Customer-facing Support Agent



Healthcare & Life Sciences

Single Cell Models to Identify Drug Targets

Next-Best-Action to Optimize Provider Engagement

Clinical Data Abstraction to Accelerate Research

Member Conversation Analytics & Insights

Claims Denial Appeals to Increase Reimbursement



Media & Entertainment

Conversational AI for Consumer Engagement

Contextualized Ad Targeting to Improve ROAS

Auto-Generate Localized Content

Autonomous Network Operations

Anomaly Detection to Prevent Fraud



Retail & Consumer Goods

Optimized Store Level Product Ordering

Agentic Commerce Search

Visual Concept Design

Hyper-Personalized Offers at Scale

Smart Inventory Optimization



Manufacturing & Auto

Imaging Defect Detection

Proactive Maintenance to Increase Uptime

Prescriptive Field Support

Process Optimization to Reduce Energy Use

Supply Chain Simulator to Mitigate Risks (e.g., tariffs)



Energy & Utilities

Predictive Power Outage Prevention

Asset Reliability to Reduce Disruption

Wildfire Risk Prevention

Hyper-personalized Energy Savings Recommendations

Regulatory Compliance Co-pilot



AI Agents demand more than BI semantics

Trusted Metrics with the same trusted definitions used for BI

Additional metadata like synonyms, LLM instructions, verified queries & snippets to enable new questions.

Both need governance, curation and management.

The screenshot displays a data management interface. At the top, a search bar and navigation icons are visible. Below, a table titled 'System Billing Genie' is shown with columns for ID, Status, Name, and Value. A modal dialog titled 'Edit column comments' is open, allowing users to add or edit comments for various columns. The dialog lists columns like 'transaction_id', 'first_name', 'last_name', 'email', 'credit_card', 'monthly_payment', 'car_make', 'car_model', 'car_model_year', and 'dealership_zip', each with a text area for a comment. To the right of the dialog, a panel titled 'About this table' provides metadata such as Owner (Kristen Wilder), Type (Managed), Data source (Delta), Popularity, Last updated (last month), and Size (64KiB, 1 file). It also includes a 'Quality' section with 'Healthy' status and 'Freshness' as 'As expected'. A 'Tags' section shows a lock icon and a heart icon. A 'Row filter' section has an 'Add filter' button. An 'Insights' section shows 'No active user in past 30 days', 'No tables joined in past 30 days', and 'No related assets in past 30 days'. At the bottom of the dialog, it says '10 comments unsaved' and has a 'Save all' button.

ID	Status	Name	Value
5	0	JUDGE_REQUEST	1
6	0	NETWORK_BYTE	13,935
7	0	STORAGE_SPACE	416
8	0	TOKEN	13
9	1	AP_OPERATION	90
10	1	COMPUTE_TIME	14,041
11	1	API_TIME	24

Edit column comments

Column	Comment
transaction_id bigint	A unique identifier for each transaction, allowing for tracking and referencing specific transactions within the dataset.
first_name string	The first name of the customer involved in the transaction, useful for personalizing communications and records.
last_name string	The last name of the customer, which, along with the first name, helps in identifying and addressing the customer accurately.
email string	The email address of the customer, which can be used for sending confirmations, updates, and promotional materials.
credit_card bigint	The credit card number associated with the transaction, necessary for processing payments and verifying customer identity.
monthly_payment double	The amount the customer is required to pay monthly, providing insight into the financial commitment associated with the transaction.
car_make string	The manufacturer of the car involved in the transaction, which can be important for inventory and customer preferences.
car_model string	The specific model of the car, allowing for detailed tracking of sales and customer interests in particular vehicles.
car_model_year bigint	The year the car model was manufactured, which is essential for understanding the age and value of the vehicle.
dealership_zip string	The ZIP code of the dealership where the transaction took place, useful for geographic analysis and market segmentation.

10 comments unsaved [Save all](#)

About this table

Owner: Kristen Wilder
Type: Managed
Data source: Delta
Popularity:
Last updated: last month
Size: 64KiB, 1 file

Quality
● Healthy
as of 4 weeks ago
Freshness: As expected
[See results](#)

Tags

Row filter
[Add filter](#)

Insights
No active user in past 30 days
No tables joined in past 30 days
No related assets in past 30 days
[All insights](#)

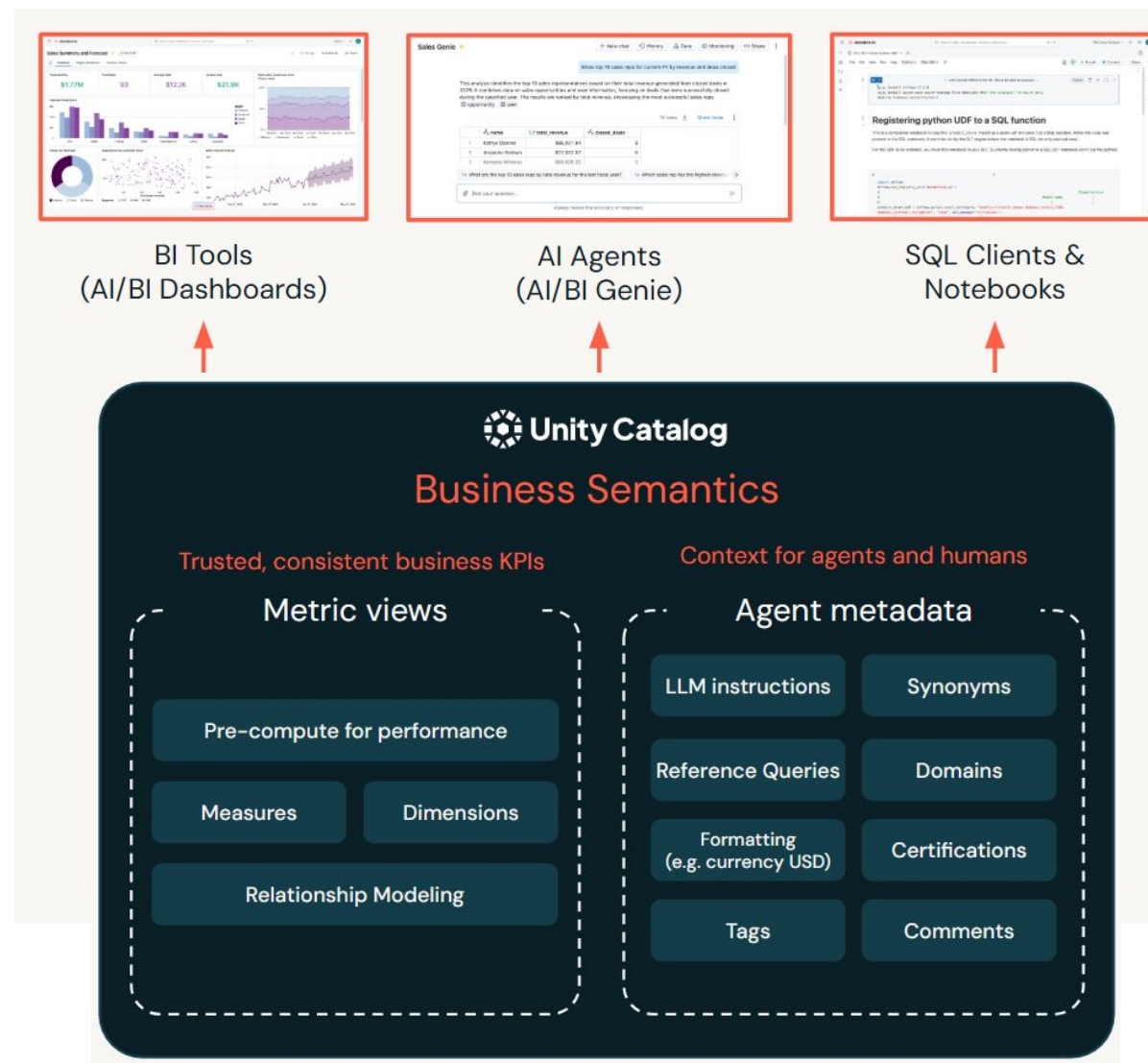


Introducing Unity Catalog Business Semantics

Open and reusable semantics across dashboards, notebooks, SQL applications, and AI agents

Establish trust and reliability through centralized governance in your data intelligence platform

AI-ready context to answer new questions accurately and adapt as business concepts evolve



How AI Can help in Metadata Generation

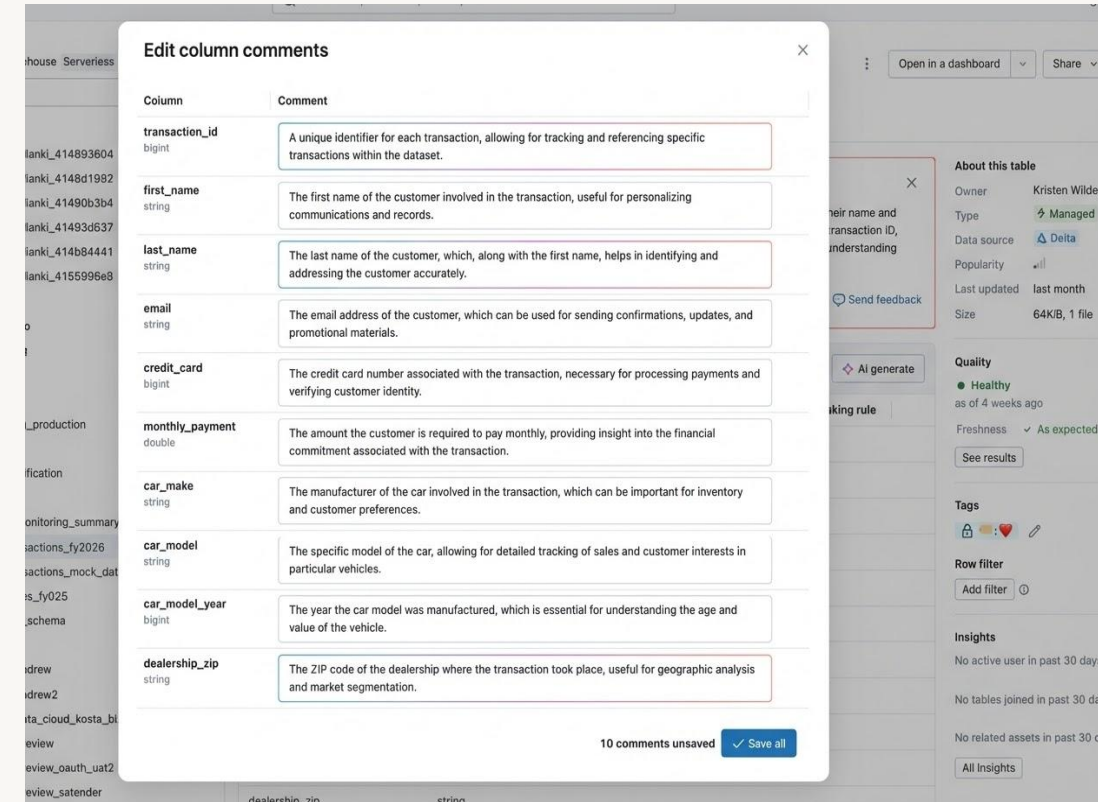


AI Powered Metadata Generation

Unity Catalog utilizes AI to generate metadata at both the column and table levels, leveraging available context such as tags, column names, and table information.

Additional metadata, including synonyms, LLM instructions, and verified queries/snippets, is generated to facilitate novel queries.

Both generated and additional metadata require robust governance, curation, and management.



What is a “customer”?

A user who
signed up via
the app

All leads in Salesforce
are potential
customers

Someone who
placed at least
one order

Other business
units are our
customers

What is a “customer”?

Active customers
placed at least one
order in the past 90
days

These suppliers
are also our
customers!

These customers have
closed their account

customer_status

order_tracking

api_logs

A user who
signed up via
the app

All leads in Salesforce
are potential
customers

Someone who
placed at least
one order

What is a “customer”?

Other business
units are our
customers

Active customers
placed at least one
order in the past 90
days

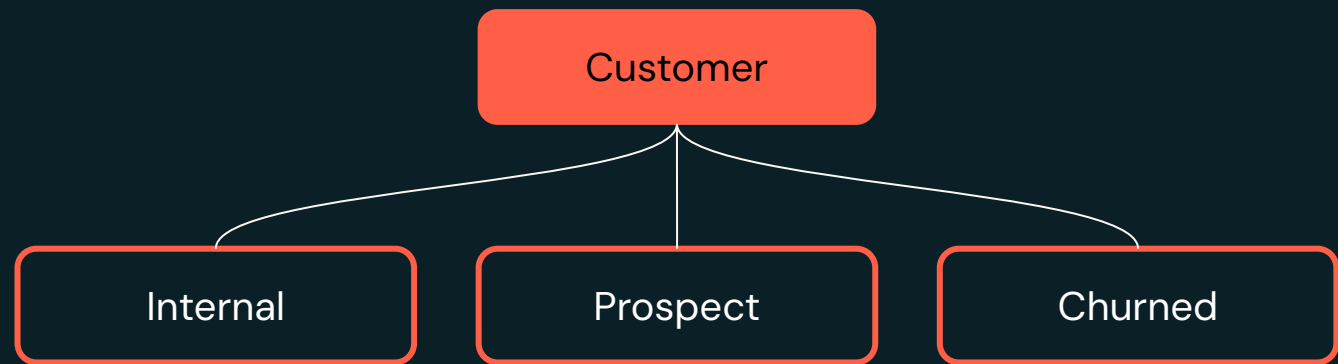
These suppliers
are also our
customers!

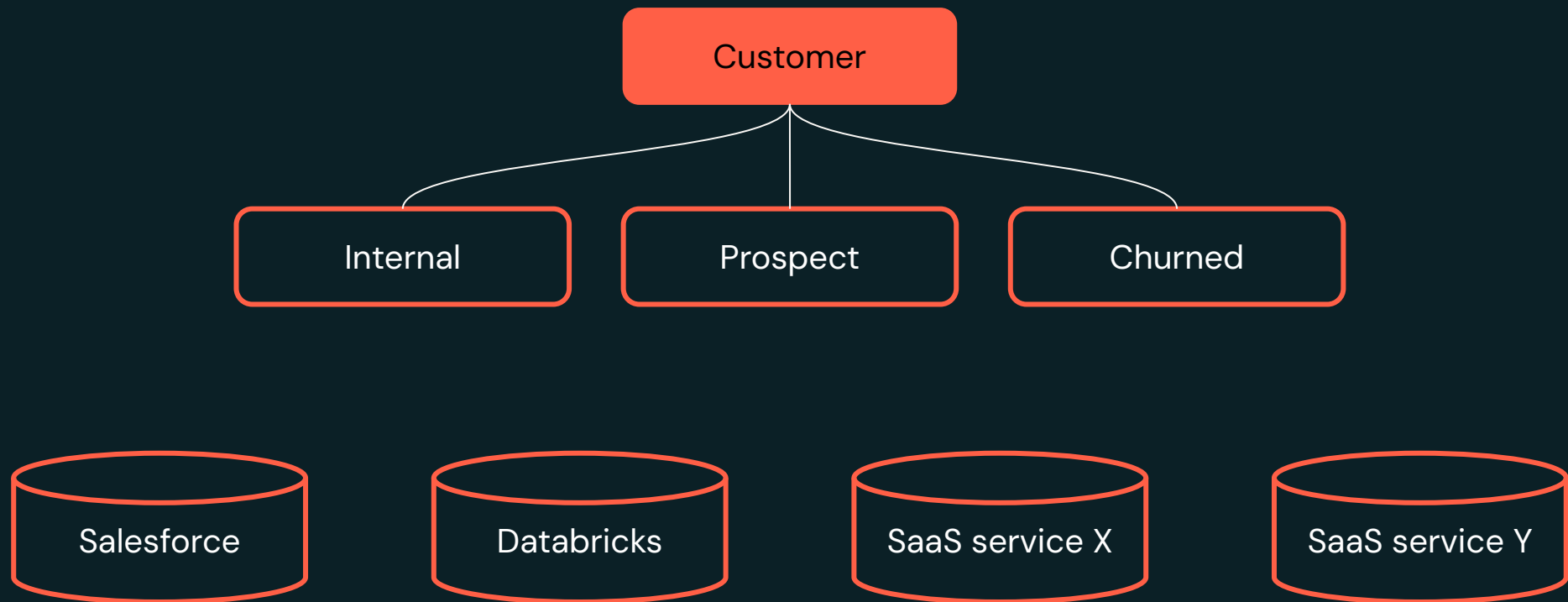
These customers have
closed their account

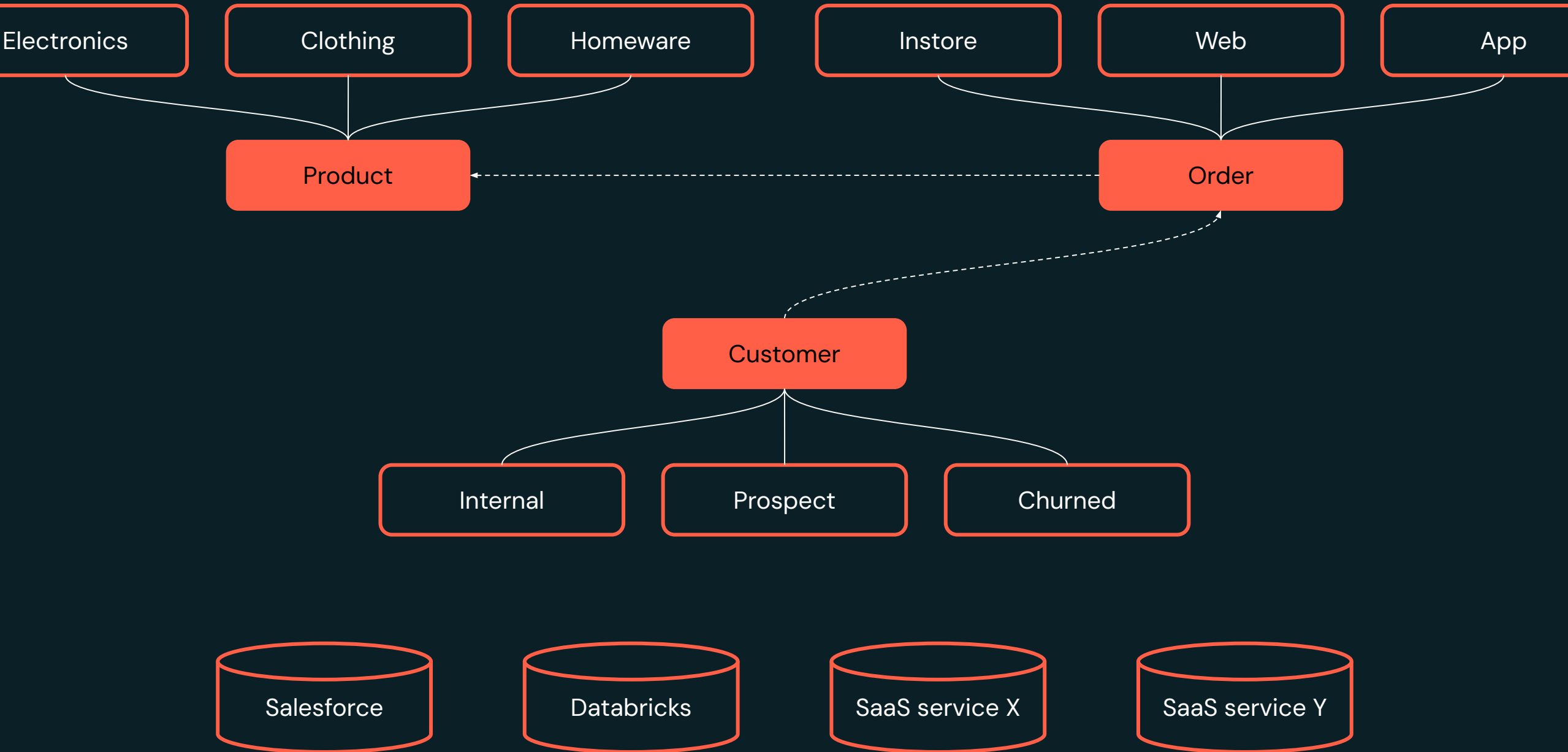
app_users

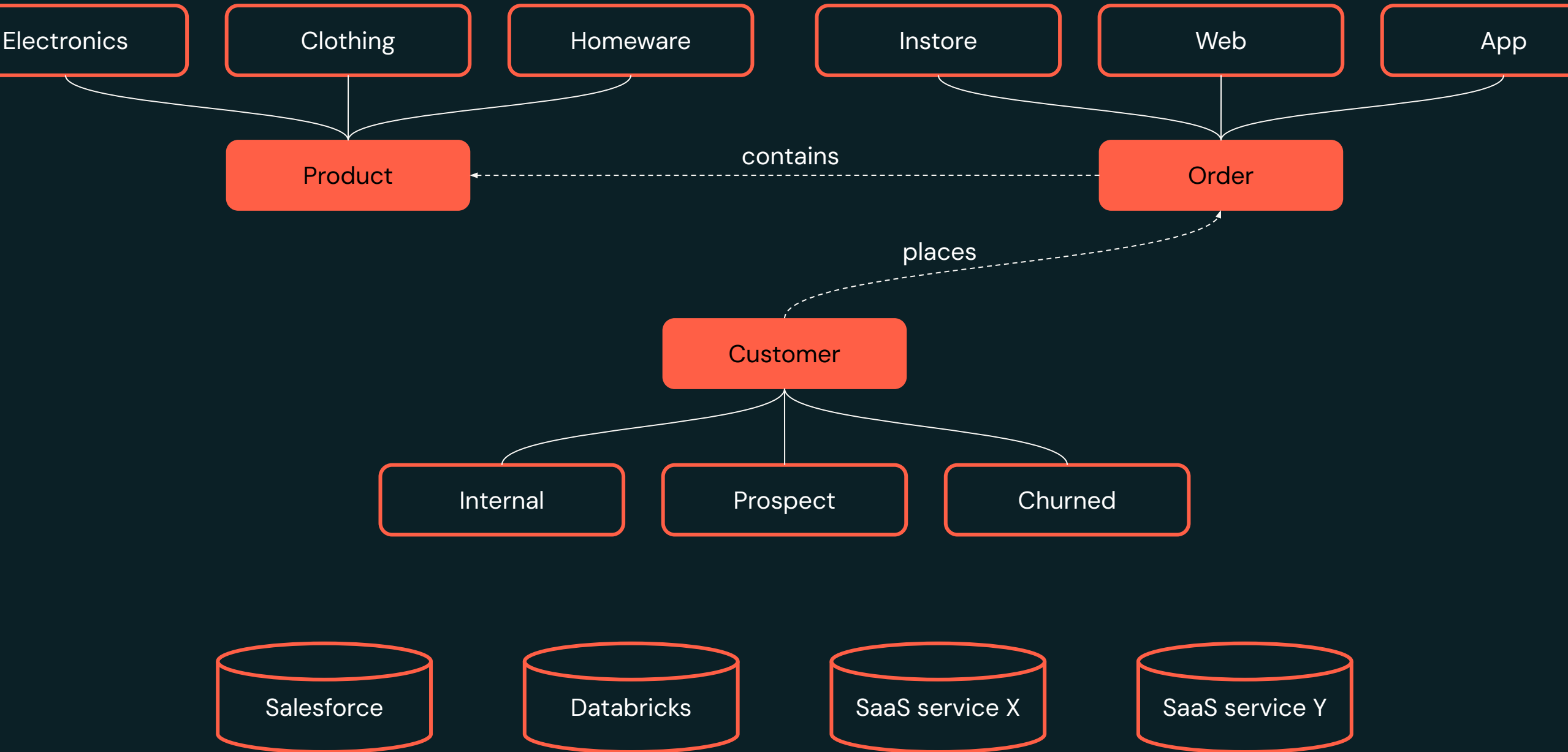
user_profiles

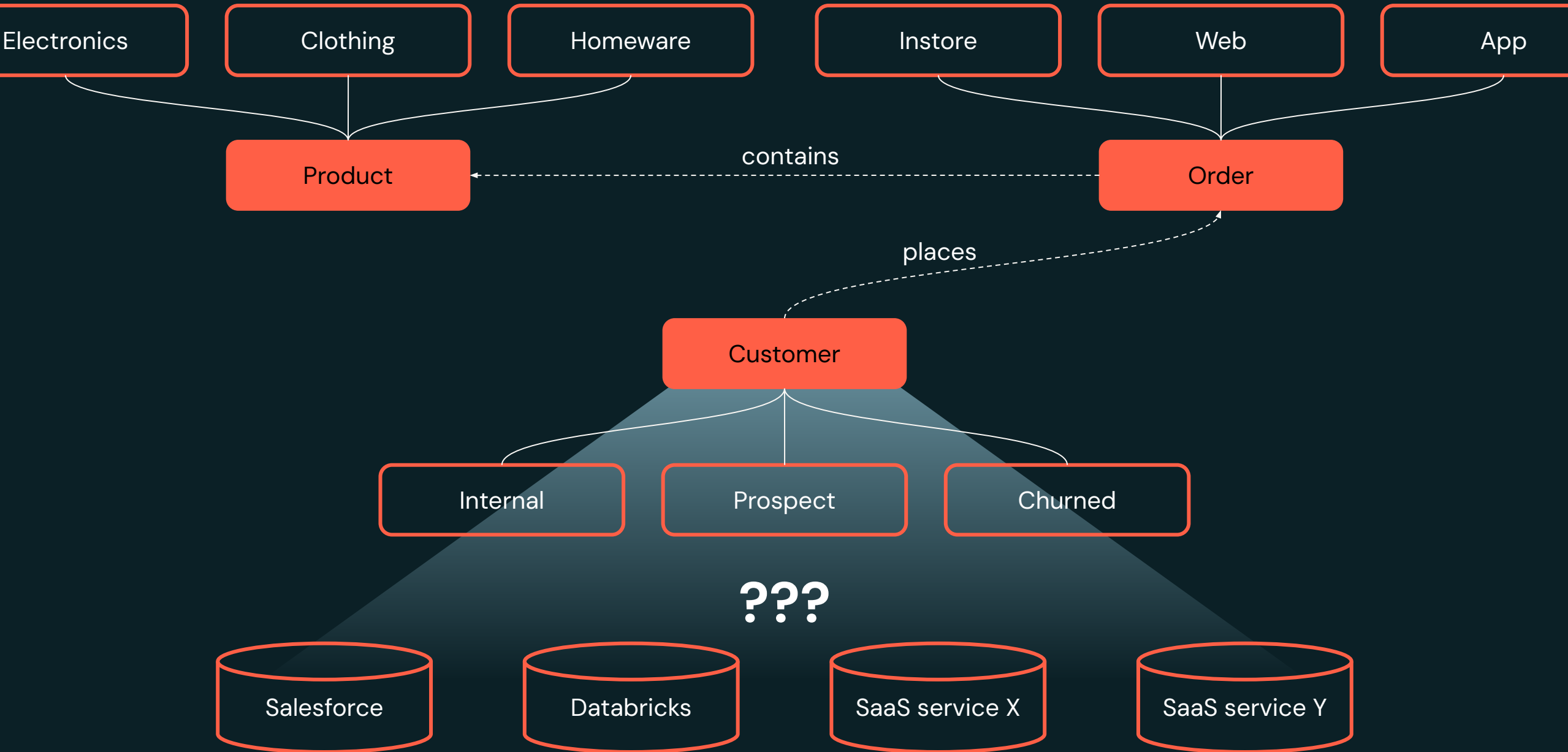
rtbf_requests_march_2024.xls











**We need a way to model the
business, not just the data.**

How AI Can help in Improving DQ.



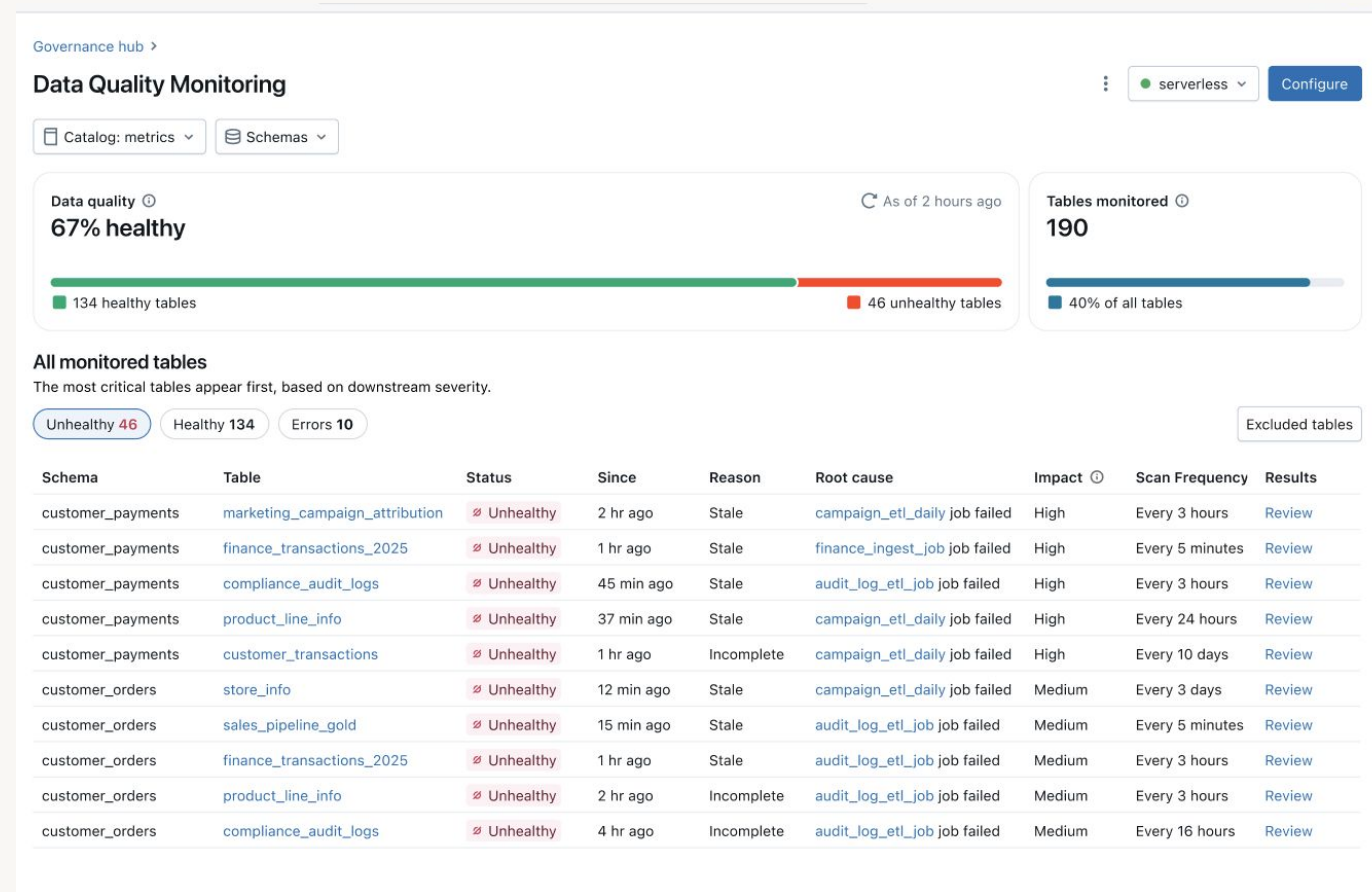
Data Quality Monitoring

Detect and resolve data quality issues at scale



Agentic Anomaly Detection

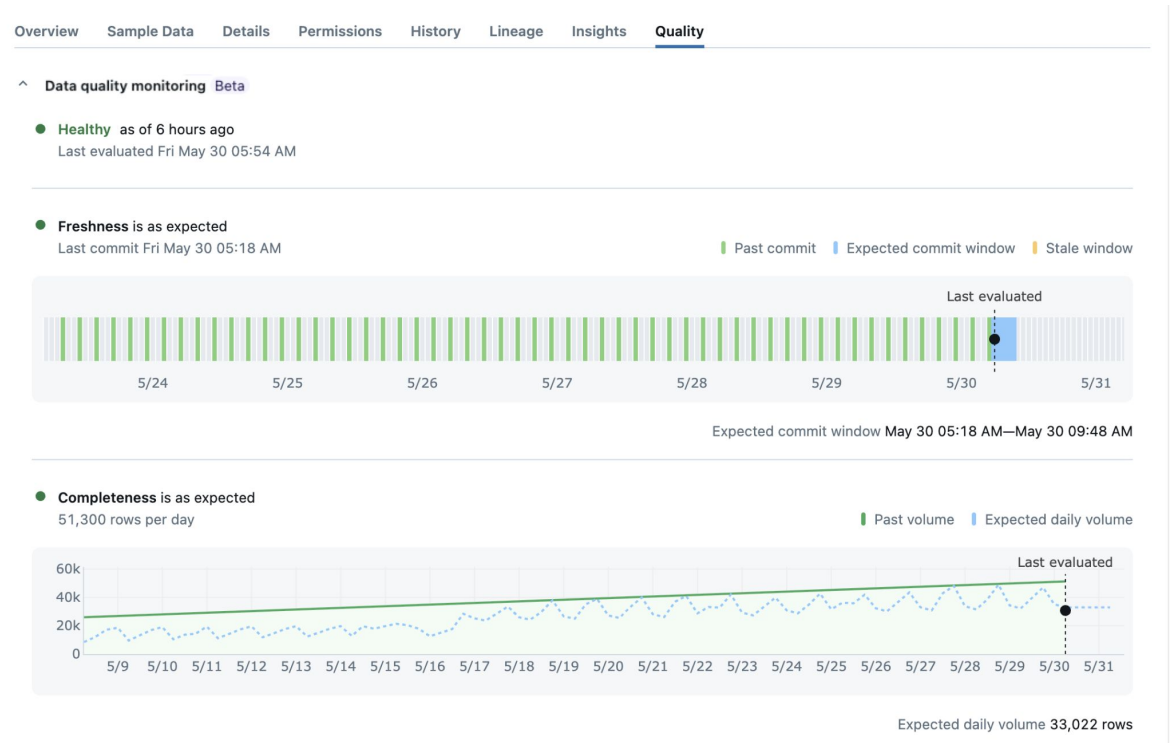
- **Learned behavior, not static rules:**
Enabled at the schema level, agents learn from historical table patterns and monitor key quality signals (Freshness and Completeness)
- **Intelligent scanning:** All tables are scanned once, then revisited based on table importance and update frequency.
- **Root cause** is surfaced directly in upstream Lakeflow jobs and pipelines, cutting time to resolution.
- **Issues are prioritized** using Unity Catalog lineage and certified tags, ensuring high-impact datasets are addressed first.



AI-Powered

Intelligent and built on your data

- **Freshness:** tracks how recently data was updated, flagging tables as stale if updates are delayed. **(Beta)**
- **Completeness:** checks if row volume in the last 24 hours falls below expected levels. **(Beta)**
- **Segmentation:** View quality metrics by slice (e.g., freshness of vendor_id="walmart") (PrPr)



~~Bespoke rules per table~~

Simple and scalable to all tables

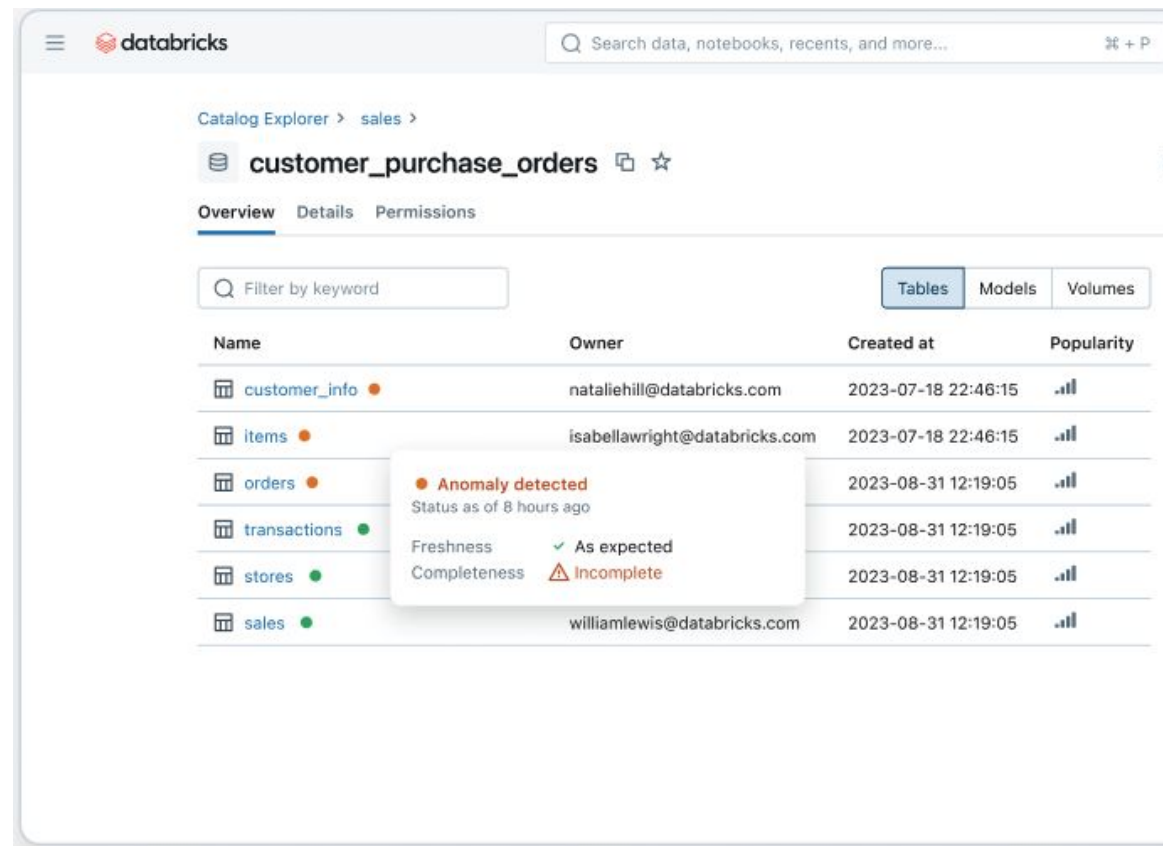


Coverage that scales

Quality understood by all

Health Indicator

- Populates each table with health indicators so data consumers can **understand if data is safe to use, at a glance.**



The screenshot shows the Databricks Catalog Explorer interface. The breadcrumb path is 'Catalog Explorer > sales >'. The selected table is 'customer_purchase_orders'. Below the table name are tabs for 'Overview', 'Details', and 'Permissions'. A search bar labeled 'Filter by keyword' is present. To the right are tabs for 'Tables', 'Models', and 'Volumes'. The main table lists several tables with columns: Name, Owner, Created at, and Popularity. A tooltip is displayed over the 'orders' table, showing an 'Anomaly detected' status as of 8 hours ago. The tooltip details 'Freshness' as 'As expected' and 'Completeness' as 'Incomplete'.

Name	Owner	Created at	Popularity
customer_info	nataliehill@databricks.com	2023-07-18 22:46:15	...
items	isabellawright@databricks.com	2023-07-18 22:46:15	...
orders		2023-08-31 12:19:05	...
transactions		2023-08-31 12:19:05	...
stores		2023-08-31 12:19:05	...
sales	williamlewis@databricks.com	2023-08-31 12:19:05	...

Anomaly detected
Status as of 8 hours ago

Freshness ✓ As expected
Completeness ⚠ Incomplete

~~Quality only understood by central team~~

Quality understood by all



DE

DE Agent & DQX



Data Quality Monitoring

Roadmap

Public Preview

- Freshness and Completeness
- Health indicators (Schema overview and Lineage)
- UI, [API](#), Terraform

February 2026

Mid 2026

Data quality by default

- Account-wide enablement and GovHub integration
- More rules (e.g., %Nulls, Uniqueness, Validity)
- More health indicators in surfaces like Genie, Dashboards, agents, etc.

Late 2026

and more...

Scalable incident resolution

- Refine table importance score and report on zombie tables
- Deeper integration with Lakeflow to enable faster RCA
- Automated Alerting

- Quarantine bad data
- Open ecosystem (plug in rules from DQX, Monte Carlo, etc.)

